Market Maker

by moontower

A market in a coin that does not exist. **You and the people at the table set its price** — and then four dice decide what it was worth.

Use `→` to move, `Esc` for the whole deck, `S` for notes. Four slides have **sliders you can drag**.

THE GAME

Three things, and that is the whole game

ONE COIN

100

It opens at a price the host sets — **100** by default — and moves every round. There is nothing else to trade.

NO MONEY

∞

You never run out of cash and you never post margin. **Buy as much as you dare.** Nothing stops you but the price.

ONE SCORE

P/L

Profit and loss is the only number kept. **Every round it is recomputed from scratch** against where the coin closed.

> Because cash is infinite, size is free and only being wrong is expensive. That is deliberate — the game is about reading a market, not about managing a bank balance.

SHAPE OF A ROUND

Five beats, in this order, every round

1 **The option auction** IF SWITCHED ON
You quote a price you would buy at and a price you would sell at. Everyone's quotes clear at once.

2 **Stock orders**
How many coins you want, and whether you want them now or only at today's price.

3 **Cards**
Everyone secretly plays cards that push the coin up, down, or wilder.

4 **Four dice**
Thrown one at a time. They decide which way the coin went and how far.

5 **Settle**
The coin lands on a new price. Everyone's P/L is restated against it.

> **Why options first?** Because an option you were just handed is a risk you may want to offset by trading the coin itself — and the coin trades in beat 2, in the same round. Run stock first and every hedge arrives a round too late, after the dice have already settled the thing you were hedging.

THE ONE THING TO GET STRAIGHT

Two prices, and they are not the same number

THE PRINT

103

Where you traded. Set in beat 2 by what the table asked for. If more people demanded coins than offered them, the print is above where the coin was.

THE MARK

124

Where it closed. Set in beat 4 by the dice. This is the number your P/L is measured against, and the number an option expiring today settles on.

> Nothing links them. The print can be well above the mark, which means everyone who bought is down; or well below, which means everyone who bought is up. You choose the print by how hard you push. You do not choose the mark at all.

BEAT 2 · STOCK ORDERS

Your orders make the print

MARKET ORDER

"Get me filled, whatever it costs." A market order always fills, in full. You find out the price afterwards.

LIMIT ORDER

"Only at today's price, not a penny worse." It sits there, and fills only if somebody needs what you are holding — and only as much as they needed.

pressure
`market buys - market sells`
who is demanding to trade right now

absorbed
`the limit orders facing that pressure`
how much of it the resting book soaked up

imbalance
`pressure - absorbed`
what was left with nobody to trade against — and the only thing that moves the price

> If every buyer is matched by a seller the coin prints exactly where it was, no matter how much volume went through. And a limit order on the same side as the pressure does not trade at all — nobody is selling at yesterday's price into a rush of buyers.

INTERACTIVE · DRAG THE SLIDERS

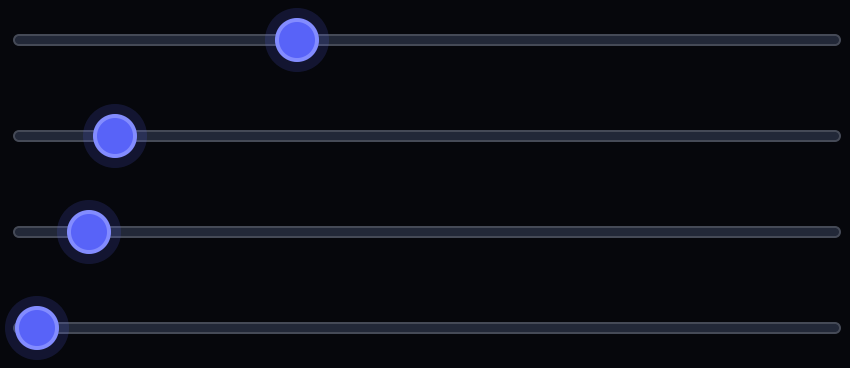
Build a round and watch the print move

Market buys
"get me in, now"

Market sells
"get me out, now"

Resting offers
sellers at today's price

Resting bids
buyers at today's price



10

3

2

0

YESTERDAY'S MARK

100

IMBALANCE

+5

THIS ROUND'S PRINT

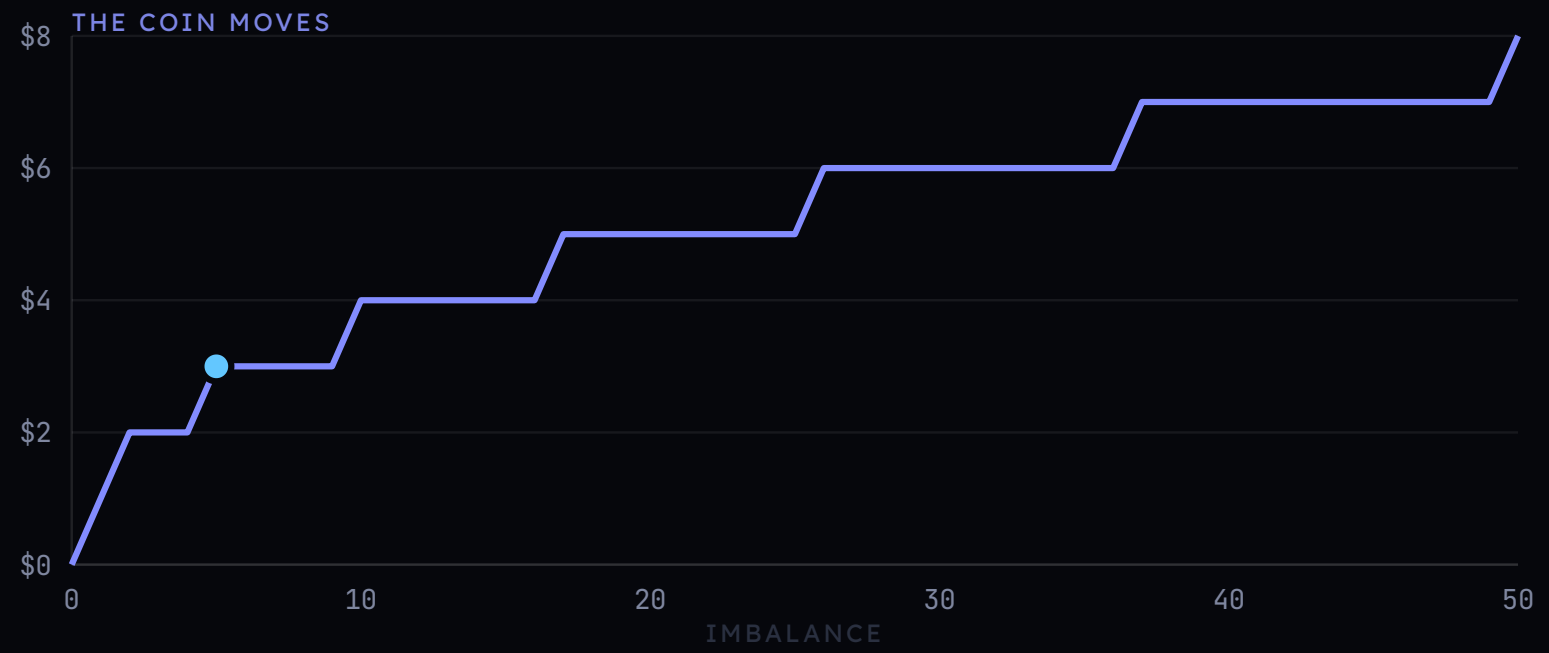
103

Pressure of **+7**, of which the resting book absorbed **2**. The leftover **5** buy orders had nobody to trade with, so the price had to move **$3** to find the other side. The house is left holding it.

> The opening numbers are the worked example: **10** bought at market, **3** sold at market, **2** resting offers soak up **2** — leaving an imbalance of **+5** and a print of **103**, three dollars above the 100 it started at.

INTERACTIVE · DRAG THE SLIDER

Pushing twice as hard does not cost twice as much



Imbalance 5

THE COIN MOVES

$3

5 coins of unmatched demand move the price **$3** — about **60¢** per coin.

Compare: an imbalance of **1** moves it **$1**. **20** — twenty times as much — moves it only **$5**.

> The move is the square root of the imbalance, **rounded up** — which is why the line climbs in steps rather than smoothly. So the first coin you demand is expensive and the fiftieth is nearly free. Size is cheap in this market — which is exactly why being on the wrong side of the dice with a lot of it is what ruins a round.

BEAT 3 · CARDS

Four kinds of card, played face down



UP

Leans the coin **upward**. Pushes the trend die.



DOWN

Leans the coin **downward**. Pushes the trend die the other way.



MULTIPLY

Makes the move **bigger**. Pushes the volatility die.



ADD

Makes the move **smaller**. Pushes the volatility die the other way.

> Only the net matters. Three ↑ against one ↓ counts as two ↑. And the net is capped by the dice: **three net cards one way settles the direction outright** — a fourth buys you nothing and is spent all the same. Everyone plays at once, in secret, so you are guessing at the table as much as at the dice.

BEAT 4 · THE DICE

Four dice, thrown one at a time

Say the table played **one net ↑ card**. That lowers the trend die's bar from 4 to **3** — and the game tells you so before anything is thrown.

TREND DIE	VOLATILITY DIE	SIZE A	SIZE B
3	**5**	**4**	**6**
3 +1 = 4 vs 3	5 vs 3	feeds the size of the move	feeds the size of the move
UP price rises	**BIG MOVE** size is A × B		

direction	up	3 plus the card cleared the bar of 3
size	4 × 6 = 24	the volatility die said multiply, not add
the coin closes at	124	100 +24

> Without that card the bar stays at 4, the **3** misses it, and the very same four dice give **-24** instead of **+24**. Which is the next slide.

INTERACTIVE · DRAG THE SLIDERS

One card can flip the whole round



THE TREND DIE NEEDS

4+

A **4** or better on one die — **50%** of rolls. No cards played: a coin flip.

THE COIN MOVES

-24

Rolled **3** against a bar of 3 → **DOWN**. Size is **4 × 6 = 24**, because the volatility die said **big move**. The coin closes at **76**.

> Start at zero cards: these dice give **-24**. Now drag one ↑ card in. Nothing about the dice changed — the bar moved from 4 to 3, the same roll now clears it, and the round is **+24** instead — **76** against **124**.
> A single card is worth 48 dollars of coin here.

BEAT 5 · SETTLE

How your score is worked out

you bought	6 coins	a market order, filled in full
you paid	$103	the print — beat 2
it closed at	$124	the mark — beat 4
your P/L	+126	(124 − 103) × 6

Every round, from scratch:
(where it closed − what you paid) × how many you hold.

Short works the same way with the sign flipped — sell 6 at the same print and you are down 126. There is no cash account, no interest, no fees. The only thing that moves your score is the gap between your price and the closing price.

THE OPTIONAL SECOND LAYER

Options

Switched on at setup. The stock-only game is exactly as it was — this adds a market in bets on where the coin ends up, and makes you the one quoting the prices.

OPTIONS · WHAT THEY ARE

Two kinds of bet, and a price you pay for it

CALL

The right to buy the coin at a number. Buy the $100 call and if the coin closes at $124 you are handed **1 coin at 100** — worth $24 more than you pay for it. Close below 100 and it simply expires.

PUT

The right to sell the coin at a number. Buy the $100 put and a close at $94 lets you sell **1 coin at 100** — a short position $6 to the good. Close above and it simply expires.

THE STRIKE

The number in the bet — $100 above. The game lists nine of them and you pick which to trade.

THE PREMIUM

What you pay for the bet. **Nobody tells you this number.** The table discovers it.

> **An option that finishes in the money turns into coin, not cash** — the position lands on your blotter at the strike and keeps moving with the market. **Nothing in this game ever quotes an option for you.** There is no model, no suggested value, no fair price on the screen. The only option price that exists is one the table agreed on — which is the entire exercise.

OPTIONS · THE BOARD

Nine strikes, re-centred every round

ORD	PX	STRIKE	PX	ORD
CALLS			PUTS	
·	—	80	—	·
·	—	85	—	?
·	—	90	—	·
·	—	95	—	·
S4	12	100	—	?
·	—	105	—	·
B?	—	110	—	·
·	—	115	—	·
·	—	120	—	·

- Calls read down the left, puts down the right, **strikes up the middle** — the layout a real trading screen uses.
- The shaded middle row is **at the money**: the strike sitting exactly where the coin is, **$100** here. Four strikes either side, **$5 apart**.
- The board **re-centres every round**. If the coin closes at **112**, next round's strikes run **92** to **132** and the old ones are gone.
- **Your position keeps its strike forever.** A 100 call you bought settles against 100 even after that strike has drifted off the board.

You quote both sides into a question mark

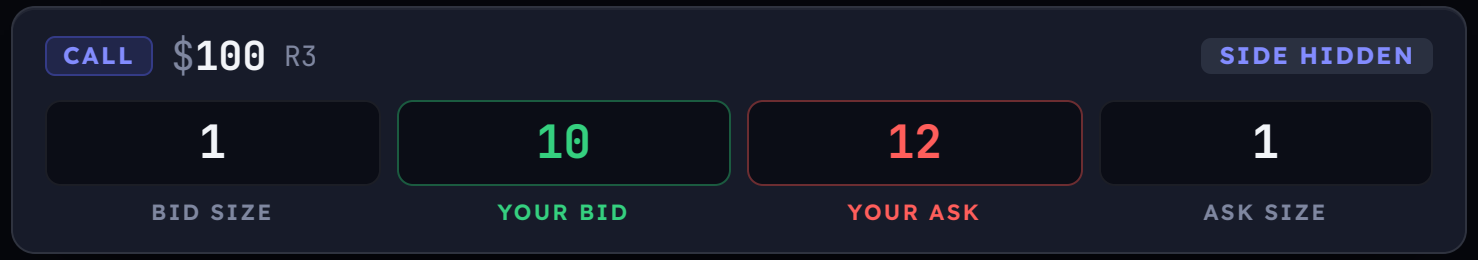
- The board shows **which strikes carry an order** — and nothing else. Where the orders landed is itself a clue.
- SIDE HIDDEN shown as **?** — you do not know whether the house is buying or selling. **You must quote a price both ways.**
- HOUSE BUYS shown as **B?** — sometimes it tells you the direction up front. Shade accordingly.
- **Size is always hidden.** Even a telegraphed order might be for one contract or for ten.

> **Quoting a market is not the same as filling an order. You have to name a price you would stand behind in either direction.**

Everybody quotes at once, in secret, and the gate stays shut until every player has confirmed or passed. Then the orders are revealed and everything clears together. **A quote you typed but never confirmed does not trade** — not even if the host forces the gate.

OPTIONS · ENTERING A QUOTE

Four boxes, reading outward from the middle



- **Your bid** is the most you would **pay**. Green, because green is what you make money on.
- **Your ask** is the least you would **accept**. Red.
- **Sizes sit outside the prices**, so a two-sided market reads out from its own spread — exactly like the board.
- Leave a side blank to quote **one way only**. Leave everything blank and confirm to **pass**.

> Only strikes carrying a house order are listed, because only those can clear. A quote anywhere else could never trade.

OPTIONS · CLEARING

Everything trades at one price

The house wants to **buy 1** of the CALL **$100**. Three people quoted.

QUOTED	THEIR BID	THEIR ASK	WHAT HAPPENED
Ana	–	12	best offer → **sold 1**
Ben	–	18	behind the best
You	20	–	bidding, not offering

the clearing price	**$12**	where the first trade happened
house bought	**1 from Ana**	at $12
then: you bought	**1 from Ben**	at $12, not his 18

> Ben asked **18** and sold at **12**. Your bid of **20** crossed his offer, and the clearing price governs every trade in the contract — so you paid 12 for something you had said you would pay 20 for, and he took 12 for something he had asked 18 for. **Offering behind the best price is not safe. It is a resting order at a price you never named.**

Two ways showing size can cost you

THE HOUSE WALKS THE BOOK

House buys **5**. Ana offers **10** for **2**, Ben offers **14** for **4**.

clears at	$10	Ana's price, first trade
Ana sells	2 @ $10	
Ben sells	3 @ $14	his own price — a better one

When the **house** runs past the best price, each further level gets **its own** limit. Being second in line pays better here.

A PLAYER CROSSES BEHIND IT

House buys **2** of the CALL $105. Ana offers **9**, Ben offers **11**, you bid **14**.

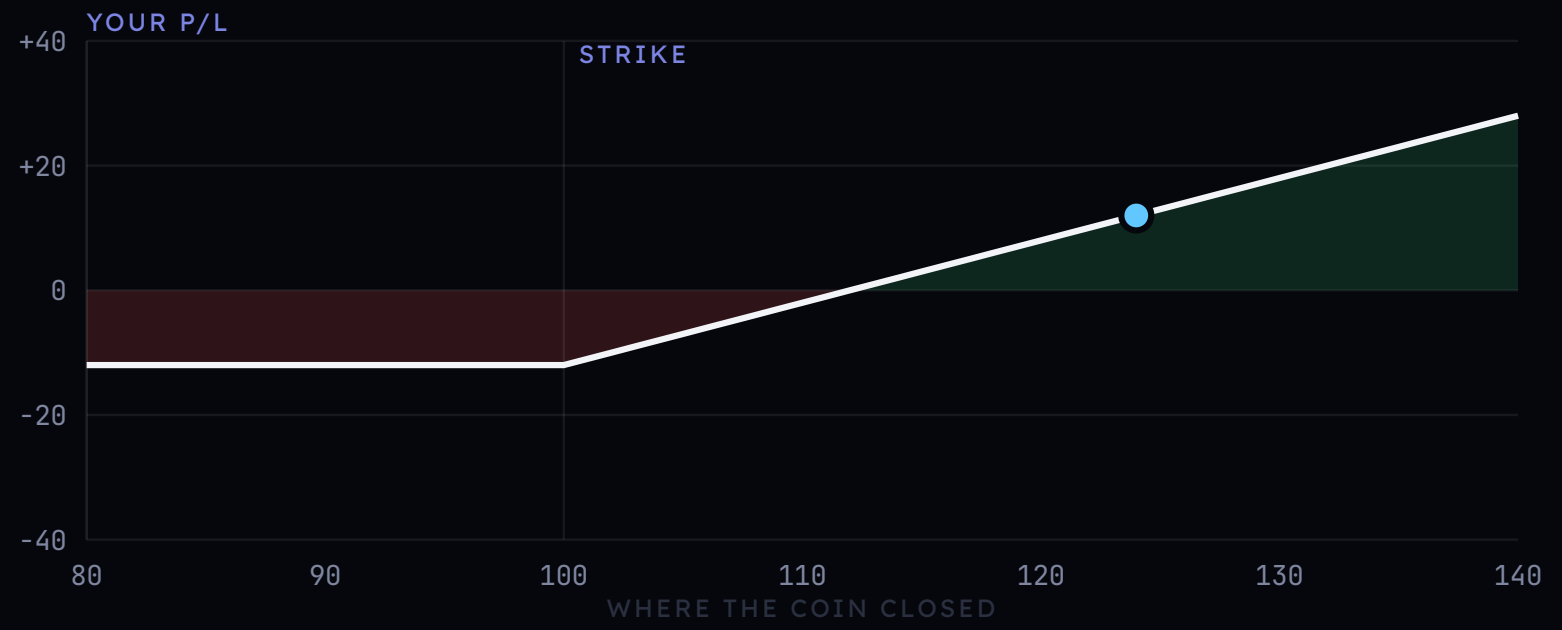
clears at	$9	Ana's price, first trade
Ana sells	2 @ $9	to the house
Ben sells	2 @ $9	to you — he asked 11

Between **players**, everything prints at the clearing price. Ben's 11 was never honoured, and your 14 was never charged.

> **And if nobody quotes the side the house needs, the contract simply does not trade.** No fill, no price, no robot stepping in to make one. A strike everybody passed on stays blank all round.

INTERACTIVE · DRAG THE SLIDER

What your option is worth when the dice land



The coin closes at $124

You are **Buying it** | Selling it

BOUGHT THE $100 CALL FOR $12

+12

You take **one coin at 100**, now worth 124: **24** to the good, less the **12** you paid. Past break-even, and from here **there is no ceiling**.

Settlement is arithmetic, not opinion: in the money, you take the coin at the strike and it is worth what it is worth; either way the premium is spent.

> Three places to stop the slider: a close at **124** makes the call worth **24** against the 12 you paid, so **+12**; a close at **112** is worth exactly what you paid, so **0**; a close at **94** is worth nothing, so **-12** — and never worse than that. Flip to selling and every number is mirrored: the seller's best round is **+12** and their worst has no floor at all.

GO

Six things worth knowing on your first game

- **Confirm, or you are not in.** Typed is not traded. Every beat is a gate and it can shut without you.
- **Passing is a move.** No order, no quote, no cards — perfectly legal, and often right.
- **Size is cheap, being wrong is not.** The print barely moves on big orders. The mark does not care how big you are.
- **Watch what the table did.** The order book is shown after every round. Who bought, how much, at what.
- **Quote wider than feels clever.** You are naming a price in both directions against an unknown. Once is enough to learn this.
- **Short options have no ceiling.** It is the one position in the game that can lose more than you expected to make.

> **The whole game is one question, asked five times:**
> **what is this worth, and is the table wrong about it?**